# DESIGN AND IMPLEMENTATION OF AN INTEGRATED FINGERPRINT BIOMETRIC AND WEB-BASED STUDENT MANAGEMENT SYSTEM

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## DRAFT STATUS

This report prototype incorporates the Chapter One, Two and Three working revisions dated 17 October 2026. Their descriptions of implemented and proposed work have been retained. It is a document-formatting and review prototype, not evidence that the complete system has been deployed or evaluated.

Chapters Four and Five are labelled completion outlines. Author details, acknowledgements, final abstract, actual results, conclusions and supporting photographs remain to be completed and reviewed. Bibliographic metadata has been transcribed from the supplied working references and requires final source checking. Internal team notes remain in the accompanying source-notes folder rather than in the academic chapters.

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## LIST OF ABBREVIATIONS

**API** Application programming interface

**CGPA** Cumulative grade-point average

**GPA** Grade-point average

**HTTP** Hypertext Transfer Protocol

**I2C** Inter-integrated circuit

**RTC** Real-time clock

**UART** Universal asynchronous receiver/transmitter

**USB** Universal Serial Bus

**UUID** Universally unique identifier

# ABSTRACT

**Abstract pending.** Write the final abstract after the implemented scope, evaluation datasets, measured results and conclusions have been reconciled. Summarise the records-management problem, integrated design, methods, principal results and limitations. No measured result has been invented for this prototype.

# CHAPTER 1

# INTRODUCTION

## 1.1 Background of the Study

Student records provide information needed for departmental administration throughout a student's academic history. These records include identifying information, academic results and documents that support administrative decisions. Their value depends on whether authorised personnel can locate the correct information, interpret it in its proper context, and update it without losing its association with the student concerned. Filing is therefore more than the physical storage of documents; it involves organising records so that they remain retrievable, updatable and usable over time.

Gill [1, pp. 3-6] describes records management through the creation, use, storage and subsequent disposition of recorded information, with retrieval serving as an important test of a filing system's usefulness. These principles provide a foundation for departmental student filing. Records must be classified and indexed, linked to the appropriate student, and handled through defined procedures. Converting a paper document into an electronic file addresses its storage format, but effec-

tive digital filing also requires suitable metadata, retrieval arrangements and controls over access and correction.

In the Department of Electrical and Electronic Engineering, Obafemi Awolowo University, the project team has observed that student records are predominantly maintained in hard-copy form. Locating the required records can be difficult, and reviewing, updating and correcting information requires considerable effort. This observation motivates the investigation of a web-based filing system that can organise student information and supporting documents within an accessible departmental application.

A digital student filing system has two related forms of information to manage. Structured records contain fields such as student identifiers, biodata, academic periods and results that can be validated, queried and processed. Uploaded documents preserve supporting material, including scanned records, and require descriptive information that makes them identifiable and retrievable. These forms should be associated with the same student record while retaining their different storage, correction and access requirements. Uploading an image of a document does not by itself populate the structured database or make every word in the image searchable.

Web-based academic information systems provide an established approach to coordinating records and user responsibilities. Aliyu [2, pp. 2-3, 20–26] describes a university academic information system with integrated data and functions for different categories of users. Such work supports consideration of a departmental application in which

the interface, application services and persistent records have defined responsibilities. A web interface can provide a common point of access, while server-side rules determine which information each user may view or change.

Records management also requires a distinction between locating a student's information and confirming the identity of a person presenting for a departmental activity. A name or matriculation number can identify a claimed record, but an identity-dependent workflow may require additional evidence that the person presenting is associated with that record. Fingerprint biometrics provides one means of obtaining such evidence through enrolment, acquisition and comparison against an enrolled reference. Recognition performance depends on the acquired sample, the reference data, the matching procedure and the operating conditions; it should therefore be evaluated rather than assumed.

Biometric links to student information are already represented in the literature. Omorogiuwa, Nwelih and Aweh [3] describe a fingerprint-based student information retrieval system, while Adetiba et al. [4] describe online biometric class attendance management. These studies establish relevant prior approaches. The present project concerns their application within a departmental system that combines the management of structured records and uploaded documents with a defined fingerprint-assisted workflow.

The project therefore brings together a web-based student filing application and a fingerprint biometric subsystem. The filing application is responsible for organising and managing records. The biometric

subsystem supplies identity evidence for selected activities, including the documented attendance workflow. Their integration must associate identity outcomes with the correct student record while preserving the permissions and rules governing that activity.

## 1.2 Statement of the Problem

Student records in the Department of Electrical and Electronic Engineering, Obafemi Awolowo University, are predominantly maintained as hard copies. As observed by the project team, these files are difficult to retrieve, and considerable effort is required to review, update and correct the information they contain. Access to the required information consequently depends on locating and handling the relevant physical records. This creates a practical records-management problem for departmental activities that require an accurate and up-to-date view of a student's information.

Addressing this problem requires more than storing electronic copies of existing papers. Structured student and academic information must be organised consistently, while uploaded documents must be indexed and associated with the correct student. Authorised users need to retrieve both forms of information and make controlled corrections without confusing students, academic periods or document versions. The first problem addressed by this project is therefore how to organise and manage these records within a coherent web-based departmental filing system.

A related problem arises in activities that require confirmation of a student's identity, such as attendance recording. The system must associate the person presenting with the appropriate student record before that identity outcome is used within the activity. Fingerprint recognition can support this association, but the recognition outcome must be integrated with the application's student identifiers, permissions and workflow rules. Capturing a fingerprint or receiving a matching result alone does not establish all the conditions required for a valid attendance record or unrestricted access to student information[1].

The project consequently addresses two connected requirements: the management of structured student records and supporting documents, and the integration of fingerprint-based identity recognition with selected departmental workflows. The engineering problem is to coordinate these functions so that records can be retrieved and updated appropriately and biometric outcomes can be linked to the correct student record under defined operating conditions.

## 1.3 Research Questions

1. How can a web-based departmental filing system be designed and implemented to organise structured student and academic records alongside uploaded documents, while supporting accurate retrieval, controlled updating and authorised access?

2. To what extent can a fingerprint biometric subsystem be in-

[1]

tegrated with that system to associate a presented fingerprint with the appropriate student record and support reliable, defined identity-dependent workflows?

## 1.4 Aim and Objectives

The aim of this project is to design and implement an integrated fingerprint biometric and web-based student management system for organising departmental student records and supporting student identification within selected departmental workflows.

The specific objectives are to:

1. Analyse the existing departmental filing workflow and establish the functional and access requirements for student records, supporting documents and the selected biometric workflow.

2. Design a data model that associates student information, academic records, document metadata and biometric identity references through stable student identifiers and appropriate integrity rules.

3. Implement a web-based application for the controlled creation, updating and retrieval of structured records, together with the upload and retrieval of supporting documents.

4. Implement the fingerprint terminal functions required for acquisition, enrolment or recognition, user feedback and preservation of capture evidence under the selected architecture.

5. Integrate biometric identity outcomes with the relevant student records and selected attendance workflow, including the handling of unresolved identities, duplicate events and interrupted processing.

6. Evaluate the records functions, biometric subsystem and their integration using defined functional, recognition, timing and reliability criteria.

## 1.5 Significance of the Study

The project is intended to provide a more organised approach to departmental student filing by associating structured information and supporting documents with the correct student record. This can support authorised personnel in locating relevant information and carrying out controlled updates and corrections. The intended benefit is directly related to the reported difficulty of working with predominantly hard-copy files.

The fingerprint subsystem adds a means of establishing an identity association within selected departmental activities. Its value depends on the correctness of that association and the rules applied afterwards. Evaluating the records and biometric functions together can show both the usefulness and practical limitations of the integrated prototype.

The study also provides engineering documentation of the records model, web application, fingerprint scanner terminal, communication arrangements and evaluation procedures. Its results can inform further

development and departmental assessment. Claims of reduced retrieval time, reduced workload or improved recognition performance will be based on the measurements actually obtained rather than assumed from the adoption of web or biometric technology.

## 1.6 Scope of the Project

The project concerns a departmental prototype comprising a web-based student filing and management application and a fingerprint biometric subsystem. Its records scope includes structured student and academic information and uploaded supporting documents. The document functions require categorisation, descriptive metadata, association with the appropriate student, controlled upload and retrieval, and defined arrangements for corrections or replacement versions.

The document scope includes scanned records, photographs and other uploaded files that support the structured student information. It avoids treating the web application as a fully fledged student information system, all student results are still a product of the pre-existing university e-portal. The prototype does not attempt to make every word in an uploaded document searchable, nor does it attempt to convert all existing departmental records into electronic form. It is instead bounded by the selected student record set, the documents actually uploaded and the implemented filing functions.

The application includes access rules for the selected departmental users. Access to a record or an administrative action depends on the

requesting user's permissions and relevant assignment or ownership rules. A fingerprint result supplies identity evidence for the selected workflow; it does not replace staff authentication or automatically grant access to academic information.

The biometric scope includes the terminal functions and identity-binding arrangements needed for the selected attendance application. Acquisition, matching, evidence transfer, identity resolution and attendance recording are distinct operations whose implementation and evaluation must be reported separately. Where deferred capture is used, an unresolved capture remains pending evidence until the required identity and workflow checks have been completed.

Evaluation covers the correctness of records and document operations, permitted and denied access, biometric decisions under the stated protocol, processing times and the integration cases actually implemented. The prototype is bounded by the selected hardware, software, record set and test conditions. Large-scale institutional deployment and biometric modalities other than fingerprint recognition are outside this prototype's scope.

## 1.7 Definition of Key Terms

**Table 1.1:** Definition of key terms

| Term | Meaning in this project |
|---|---|
| Student filing system | The organisation and controlled management of structured student records and uploaded supporting documents. |
| Structured record | Information held in defined database fields and relationships, such as student identifiers, academic periods and result entries. |
| Supporting document | An uploaded digital file, including a scanned record, linked to a student and described by metadata used for filing and retrieval. |
| Student registration | Creation or maintenance of a student's departmental record. It is distinct from course registration, account creation and biometric enrolment. |
| Biometric enrolment | Acquisition and storage of a fingerprint reference and its association with the appropriate student identity. |
| Fingerprint verification | Comparison of a new fingerprint sample against a reference for a claimed identity, commonly a 1:1 operation. |
| Fingerprint identification | Search of a new fingerprint sample against an enrolled reference gallery to resolve an identity, commonly a 1:N operation. |
| Authorisation | The application's decision about whether an authenticated user may perform a particular action on a record or other resource. |
| Pending capture | Saved fingerprint evidence that has not yet completed the identity, timing and workflow checks required for an attendance decision. |

# CHAPTER 2

# LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

## 2.1 Introduction

This chapter examines the principles and previous systems relevant to the organisation of departmental student records and the integration of fingerprint identification into student-management activities. The review begins with filing, records management and information retrieval because these establish the purpose of the application. It then considers database integrity, web architecture and access control before examining biometric recognition, terminal hardware and related implementations.

The literature is assessed in relation to two requirements established in Chapter One. The first is the management of structured student and academic information alongside uploaded supporting documents. The second is the association of a fingerprint recognition outcome with the correct student record within a defined departmental workflow. This sequence connects the administrative problem to the engineering design and provides a basis for evaluating the integrated system.

## 2.2 Filing Records Management and Information Retrieval

### 2.2.1 The purpose of a filing system

A filing system organises records so that the information required for an activity can be located and used. Storage is one part of this process. Classification, indexing, access arrangements and procedures for updating or replacing records influence whether the stored material remains useful. Gill [1, pp. 3-6] treats recorded information as an organisational resource and relates effective management to its availability for subsequent use.

In departmental student administration, a student's file may contain identifying information, academic records and documents produced at different times. These items need a consistent association with the student and sufficient context to distinguish academic periods, record categories and revisions. A collection of digital files without an agreed index can reproduce the retrieval difficulties of an unorganised paper collection. The design problem therefore concerns the organisation of information as well as its conversion to electronic form.

### 2.2.2 The records lifecycle

Gill [1] describes a records lifecycle extending through creation, use, storage, retention and eventual disposition. Applied to student filing, this perspective raises questions about how a record enters the system, who may use it, how a correction is authorised, and what happens when it is superseded or becomes inactive. A record's administrative

history can remain important after its immediate use has ended.

The lifecycle perspective informs the proposed system in several ways. New records require an identified source and an appropriate student association. Corrections need an authorised procedure that preserves the meaning of the earlier information. Supporting documents require a policy for replacement, version history and retention. These are project design requirements; the detailed retention periods and disposal authority must be established for the department rather than inferred from a general management text.

### 2.2.3 Structured records and supporting documents

Structured student records contain defined fields and relationships that can be validated and queried. Examples include a student identifier, admission information, an academic session, a course offering and a result. Uploaded documents preserve supporting material as digital files. Their contents may be visible to a reader while remaining unavailable to field-based database queries.

These two forms of information require a common identity link but different processing. A result value can be checked against an allowed numerical range, while a scanned result sheet requires file validation, descriptive metadata and controlled retrieval. An uploaded scan does not automatically become verified structured data. Optical character recognition and source-data extraction would require their own methods and accuracy evaluation if included.

For this project, the conceptual student file consists of the student's

structured record and the supporting documents associated with it. Document metadata can identify the category, title, academic period, uploader, upload time and revision relationship. The file itself can be held in private storage while the application controls access through its metadata and permissions. The exact implementation of this document subsystem is developed in Chapter Three as a proposed extension to the current records backend.

## 2.3 Web Based Student Information Management

### 2.3.1 Student and academic information systems

Student information systems coordinate data used in academic administration. Aliyu [2] presents WeBAIS as an academic information system connecting student registration, result processing and functions for departmental and other university users. Its organisation around user categories and academic processes illustrates the need to design an information system around institutional activities rather than a single data-entry form.

Mardaniyan [5] similarly describes a web-based student information management design incorporating student profiles, academic structure, assessment information, search and reporting. Its emphasis on a coherent data model and differentiated user responsibilities is relevant to the present project's filing functions — a three-tier architecture of Administrator, Faculty staff, and Student. The scale, modules and evaluation of these systems differ from the departmental prototype, so

their outcomes must be interpreted within their stated contexts.

The present system adopts departmental filing as its foundation. Academic records, supporting documents and selected attendance information belong to related workflows, but they have different meanings and correction rules. A student's identity should remain stable when an external identifier changes or additional records are added. This requirement supports the use of a permanent internal student identifier rather than a person's name, a scanner slot or a mutable document label as the organising key.

### 2.3.2 Separation of application responsibilities

A web-based application can separate presentation, application processing and persistent storage. The browser presents information and collects user input. Backend services validate requests, apply the relevant permissions and execute record operations. The database preserves structured records and their relationships. Supporting documents add a private file-storage responsibility that must be coordinated with database metadata.

This separation is important for student filing because a visible interface restriction is insufficient to determine access. A request to retrieve or modify a record must be checked by the application even when it is submitted outside the normal page sequence. Similarly, a document should not become publicly retrievable merely because its storage path is known. These are design requirements of the proposed application, and their implementation must be demonstrated through

direct access tests.

### 2.3.3 Data integrity and academic history

Relational integrity supports the association of each record with the appropriate student, course and academic period. The PostgreSQL documentation describes primary keys, unique constraints and foreign keys as mechanisms for identifying rows, preventing prohibited duplication and maintaining references between tables [6]. Transactions provide a means of applying a related group of database changes as one operation [7].

The project applies these principles through student identifiers, session-specific course offerings and constraints on result attempts and attendance records. A missing score has a different meaning from a recorded zero. A change to a course's current credit units should not silently change the historical meaning of a saved result. A corrected result can also affect an academic summary or a previous review. Consequently, data integrity involves the meaning and provenance of records as well as the validity of their individual fields.

These considerations support explicit result verification, preservation of historical grading references and controlled corrections. They also explain why academic calculations and completeness decisions require independent evaluation. A database can contain internally valid rows while still lacking part of a student's academic history.

## 2.4 Access Control and Identity Responsibilities

Role-based access control associates permitted operations with organisational roles. Jansen [8] describes a formal RBAC model in which roles mediate access to resources. This provides a foundation for separating responsibilities such as administration, advising and student access.

The project requires additional context around those roles. An adviser's access depends on the assigned level and current academic session, while attendance operations depend on a course assignment. A student account is associated with its own student record. The resulting policy combines role and action permissions with the scope of the relevant record or activity.

Biometric recognition has a different responsibility. It contributes evidence about the identity of the person presenting a fingerprint. Authorisation determines whether the requesting user may view or change a protected record, and whether a recognised student is eligible for the selected activity. A successful fingerprint comparison cannot substitute for those decisions. This distinction prevents the identity subsystem from becoming an unintended route around staff or student permissions.

## 2.5 Fingerprint Biometric Recognition

### 2.5.1 Acquisition, enrolment and representation

A fingerprint recognition system acquires a sample, derives a representation and compares it with enrolled reference information. Wayman et al. [9, Chapter 1] describe biometric systems through the responsibilities of acquisition, transmission, processing, decision and storage. This framework is useful because the responsibilities can be distributed between the sensor, terminal and application without changing their conceptual purpose.

Bhanu and Tan [10, pp. 7-11] discuss fingerprint features, recognition modes and the difficulty introduced by variation between impressions of the same finger. Minutiae such as ridge endings and bifurcations provide examples of features used in fingerprint recognition. Capture position, pressure, partial contact and image quality can affect the available representation. The relative persistence of fingerprint structure therefore does not imply that every acquired sample will be identical or that every recognition decision will be correct.

Enrolment associates a reference with an identity. The correctness of this association is essential: accurate matching against a reference cannot repair an incorrect student binding created during enrolment. The system must distinguish the permanent student identity, an enrolment/reference identity, the physical scanner and the current reference location in a scanner gallery.

### 2.5.2 Verification and identification

Verification compares a new sample with the reference for a claimed identity, commonly described as a one-to-one or 1:1 operation. Identification searches an enrolled gallery to resolve an identity, commonly described as one-to-many or 1:N. Bhanu and Tan [10] distinguish these operating modes through their inputs and recognition tasks.

For a matcher in which larger scores indicate greater similarity, a simplified verification rule is to accept a comparison when its score is at least the selected threshold. A conceptual identification system selects a candidate from the gallery and applies its decision rules to that candidate. This description does not establish the proprietary algorithm inside the project's scanner or imply that the application has access to all internal comparison scores.

Fiumara [11] explains the trade-off between false matches and false non-matches and distinguishes verification from identification errors. The project must therefore state its recognition mode, reference-gallery size and trial definitions. A raw sensor matching score is supplementary information about a comparison; it is not a calibrated probability that the person has been correctly identified.

### 2.5.3 Reference compatibility and identity binding

A representation that can be exported and restored is not necessarily interoperable with a different scanner or software matcher. Wayman et al. [9] discuss the relationship between biometric processing, storage

and system interoperability. Compatibility depends on the representation and supported comparison procedure, not only on the transport interface.

The project currently labels its exported scanner representations as opaque. This avoids treating the UART packet format as proof of an ISO or ANSI minutiae format. A hash can detect equality or corruption of saved bytes, while a biometric comparison is required to relate different captures of a finger. Scanner slots and template hashes consequently cannot replace the permanent student identifier.

## 2.6 Embedded Terminal and Communication Principles

The terminal coordinates fingerprint acquisition, feedback and preservation or transfer of evidence. The ESP32 device family supplies processing and peripheral interfaces suitable for this role, including serial communication and wireless capability [12]. The availability of Wi-Fi is a component capability; it does not establish that a complete network integration has been implemented.

The tested terminal uses a regular ESP32 identified in the hardware records as ESP32-D0WD-V3. The fingerprint module communicates through UART using the compatible scanner protocol. The internal recognition functions are supplied by the module. The firmware's contribution concerns command sequencing, bounded communication, state handling, recording and integration with the wider application.

Timekeeping is a separate responsibility. The DS1307 maintains

calendar information through seconds, minutes, hours and date fields and communicates through I2C [13]. Calendar time and elapsed-millisecond timers serve different purposes. Calendar time can locate an event in an academic activity; elapsed timers measure processing intervals. Neither a timestamp nor a checksum authenticates the device that produced a record.

Network interruptions motivate separation between capture and later application processing. A terminal can preserve an unidentified capture for subsequent transfer, but that file remains evidence requiring identity and timing review. Repeated transfers should not create repeated application effects. This requirement is addressed in the project's proposed integration through stable event identifiers, immutable capture data and transactionally controlled processing.

## 2.7 Review of Related Systems

### 2.7.1 Academic records and web based administration

Aliyu [2] developed WeBAIS for Nigerian universities, with functions covering admission, registration, academic results and different staff responsibilities. The thesis is relevant to the organisation of departmental modules and the separation of user privileges. It also illustrates the distinction between implementation descriptions and appraisal evidence. Reporting inconsistencies in the questionnaire totals limit numerical interpretation, and user opinions about efficiency or security are not treated as controlled measurements of those properties.

Mardaniyan [5] presents a student information management design supported by a synthetic prototype benchmark. The paper reports operations on 2,000 synthetic records using local loopback requests, with mean timings around 0.70–1.08 ms for the operations shown. This provides an example of separating functional verification from a controlled performance experiment. Its measurements describe one local setup and cannot be treated as Internet response times or expected results for this application.

Together, these studies support the importance of an integrated records model, differentiated user access and evaluation of actual record operations. They do not remove the need to establish this department's record categories, document-management requirements and correction procedures.

### 2.7.2 Biometric student information retrieval

Omorogiuwa, Nwelih and Aweh [3] describe a biometric student information retrieval system using fingerprint recognition. Their work connects a student's biometric input with stored student information and reports a Visual Basic implementation with a student data dictionary. It is directly relevant to the transition from the filing problem to identity-supported retrieval.

The article presents system structure and application screens, but does not provide a reproducible biometric confusion matrix or a controlled retrieval-time comparison. It therefore supports the existence and design of such a workflow rather than a particular recognition

accuracy or quantified time saving. Its relatively simple data structure also differs from a design that separates academic attempts, supporting documents, scoped permissions and biometric references. These differences identify requirements for the present project without establishing that the general idea of biometric retrieval is new.

### 2.7.3 Online and desktop biometric attendance

Adetiba et al. [4] developed OBCAMS, an online biometric class attendance management system with student, lecturer and administrator functions. The architecture combines a fingerprint device, client computers, a server application and a relational database. Its evaluation used 60 students over five experiments and reported average first-signing-attempt success of 89.33%, with further attempts tabulated separately.

OBCAMS is important prior work because it demonstrates online biometric attendance and differentiated web interaction well before the present project. Its first-attempt result should be interpreted within its repeated-signing protocol. It is not a false-acceptance estimate, and the participating group should not be presented as a large set of independent identities merely because several experiments were conducted.

Olasupo et al. [14] describe a fingerprint-based attendance system for Olabisi Onabanjo University using a DigitalPersona scanner, C# Windows Forms and Microsoft SQL Server. The published architecture places the application on staff computers and the database on a

networked server. The paper provides a useful comparison for attendance functions and user interfaces. Evidence is principally presented through implementation descriptions and application screens, so a general recognition error rate is not inferred from the availability of those features.

Osasuyi, Oghogho and Ihedi-Okonkwo [15] describe enrolment, fingerprint matching, attendance recording and report generation. Their paper identifies Java and MySQL, with HTML/CSS among the interface technologies, although some platform terminology in the text is inconsistent. It reports a registration operation within approximately 80 seconds and attendance collection below 10 seconds. These bounds lack the detailed timing distribution needed for a direct comparison with the present terminal's command-level timing.

### 2.7.4 Embedded and recent hybrid implementations

Odulaja [16] developed a fingerprint attendance system using an ATmega328P controller and RS305 scanner modules. The study reports a 100-student evaluation, approximately 95% recognition success and an average verification/identification time of about seven seconds. It is relevant to the coordination of embedded hardware, user information and attendance processing.

The published round-level results and discussion do not supply a complete trial-by-trial confusion matrix from which every recognition error can be reconstructed. Its comparison of attendance across different semesters also does not isolate a causal effect of biometric adoption. The

reported results can be cited as the author's findings, while avoiding unsupported conclusions about guaranteed performance, unlimited class capacity or elimination of manipulation.

Adediran, Adetutu and Semilore [17] present OmniAttend, which combines QR-code and fingerprint pathways through a React interface, Firebase Firestore and a local Node.js bridge for a DigitalPersona scanner. The paper illustrates integration of multiple identity pathways with common attendance records. It also states limitations involving Windows support, continuous Internet dependence and a narrow separation margin in perceptual-hash fingerprint comparisons.

Its similarity threshold of 0.80 is a decision setting, not evidence of 80% recognition accuracy. The description of sub-ten-second matching should also be considered alongside the insufficiently specified participant counts and error-rate experiment. The paper is useful for architectural comparison and its reported limitations, but its matching procedure is not assumed to be suitable for the opaque representations generated by the project's scanner.

## 2.8 Comparison and Synthesis of Related Work

Table 2.1 compares the studies according to the workflow they address and the evidence relevant to this project. A feature not evaluated or described in a paper is treated as unreported, rather than automatically absent from its implementation.

**Table 2.1:** Comparison of related student information and biometric systems

| Study | Main workflow | Architecture or technology reported | Evaluation useful to this project | Implication for the present design |
|---|---|---|---|---|
| Aliyu [2] | University academic information management | Web application, relational records and differentiated users | Implementation and user appraisal; numerical appraisal requires caution | Organise functions around academic processes and permissions. |
| Mardaniyan [5] | Student and academic record management | Web design with a local relational prototype | Synthetic functional tests and loopback timings | State the dataset and distinguish server timing from user-task timing. |
| Omorogiuwa et al. [3] | Biometric student information retrieval | Visual Basic application and student information database | Functional design and screens; limited quantitative recognition evidence | Treat identity binding and retrieval as connected, separately testable operations. |
| Adetiba et al. [4] | Online biometric attendance | Web application, scanner and SQL Server | First-attempt signing outcome over repeated experiments | Account for retry policy and gallery/test population. |

Continued on next page

| Study | Main workflow | Architecture or technology reported | Evaluation useful to this project | Implication for the present design |
|---|---|---|---|---|
| Osasuyi et al. [15] | Fingerprint attendance and reports | Java/MySQL with reported interface components | Bounds on registration and attendance time | Define complete-operation timing boundaries before comparing systems. |
| Olasupo et al. [14] | Fingerprint attendance administration | Digital Persona, C# Windows Forms and SQL Server | Features and implementation screens | Separate desktop access architecture from a browser-based application. |
| Odulaja [16] | Embedded fingerprint attendance | ATmega328P, RS305 modules and computer database | Reported recognition and processing-time results | Record per-trial outcomes and avoid importing another prototype's performance. |
| Adediran et al. [17] | Hybrid QR/fingerprint attendance | React, Firestore and Node.js bridge | Functional integration and stated matching/network limitations | Evaluate the chosen representation, decision method and interruption handling. |

Three findings emerge from this comparison. First, student-information management requires a coherent records model and controlled operations even where biometrics is absent. Second, fingerprint retrieval and online attendance are established application classes. Third, performance claims depend on the particular recognition mode, dataset, retry policy and timing boundary used in each study.

The proposed system therefore requires evaluation at more than one level. Record correctness and document retrieval address the filing problem. Recognition experiments address the biometric subsystem. End-to-end tests establish whether identity outcomes reach the appropriate record and produce only the application effects allowed by the workflow.

## 2.9 Theoretical and Conceptual Framework

The framework combines records-lifecycle principles, relational integrity, scoped access control, biometric pattern comparison and reliable event processing. These principles explain different parts of the system rather than serving as interchangeable descriptions of its purpose.

**Table 2.2:** Relationship between principles design and evaluation

| **Principle** | **Application to the project** | **Evaluation consequence** |
|---|---|---|
| Records lifecycle and retrieval | Classify student information and documents; retain the context of creation, correction and replacement | Test retrieval tasks, metadata association and document-history behaviour. |
| Relational integrity | Link results and application events to stable student and course identifiers | Test duplicates, invalid references, historical context and atomic updates. |
| Roles permissions and record scope | Restrict actions by user permission and adviser/course/ownership context | Test permitted and prohibited direct requests as well as interface behaviour. |
| Biometric pattern comparison | Compare a presented sample with an enrolled reference or gallery | State recognition mode, reference set, outcomes and acquisition/processing errors. |
| Stable event identity and controlled processing | Preserve captures through transfer and prevent repeated submissions from producing repeated attendance effects | Test replay, conflict, interruption and reconciliation paths. |

Figure 2.1 shows the logical relationship between these responsibilities. It is a conceptual design rather than a claim that all components have already been integrated. Matching may use compatible scanner-assisted processing; software-only server matching is not assumed.

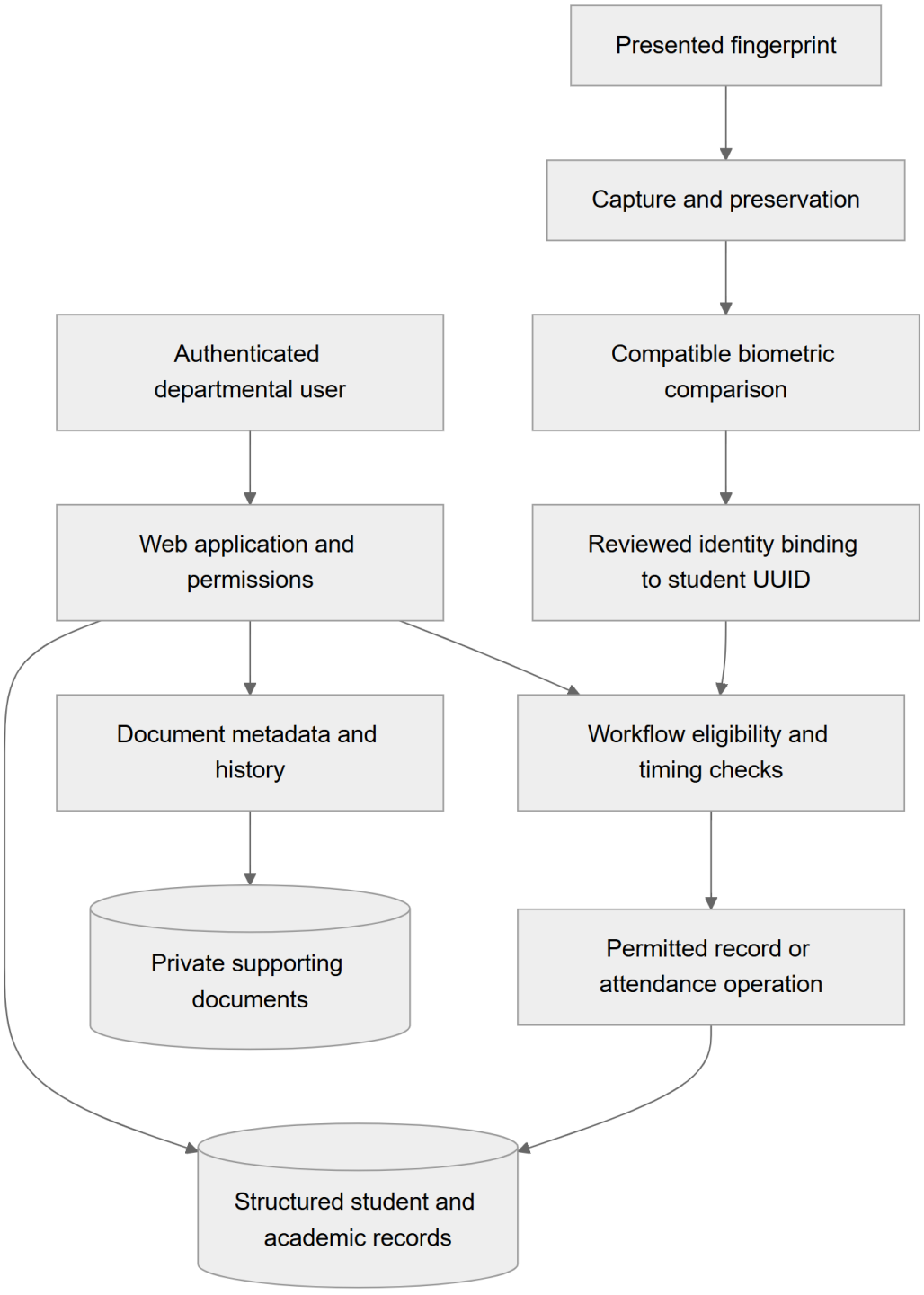



**Figure 2.1:** Conceptual relationship between filing identity and authorised operations

The student UUID provides the connection between the records and identity domains. Document storage does not depend on using a fingerprint as a filename or primary key. Likewise, a successful comparison does not automatically expose documents or academic results. The application uses the identity outcome within the permissions and requirements of the selected activity.

## 2.10 Research Gap and Project Contribution

The departmental problem reported for this project concerns predominantly hard-copy records that are difficult to retrieve, review, update and correct. The proposed scope includes structured information and uploaded documents. The reviewed literature supplies relevant records-management, academic-information and biometric-integration approaches, but the departmental combination of requirements still needs to be translated into an implemented and evaluated design.

The contribution is therefore framed as the design, implementation and evaluation of an integrated departmental system. Its assessment centres on consistent student identification, controlled records and document operations, appropriate user scope, and the handling of biometric evidence within the selected attendance workflow. The currently developed offline capture and pending-import path also creates a specific integration question: how to preserve unidentified evidence and resolve it later without creating premature attendance, false absences

or repeated records.

This is an applied engineering contribution. Any claim that the system is the first to combine fingerprint recognition with a web application, or that it outperforms the reviewed systems, would require evidence beyond the present comparison. The design's value is established by how well it meets the stated departmental requirements and by the limits shown in its evaluation.

## 2.11 Chapter Summary

The literature establishes filing and records management as the foundation of the application, with database integrity and access rules governing its operations. Fingerprint recognition supplies identity evidence that must be associated with a stable student record and interpreted within a defined workflow. Previous systems demonstrate relevant approaches but use different architectures and evaluation procedures. Chapter Three translates these principles and departmental requirements into the records model, document design, terminal configuration, integration procedure and evaluation method.

# CHAPTER 3

# METHODOLOGY AND SYSTEM DESIGN

## 3.1 Introduction

The project adopts an engineering design and development approach with iterative prototyping and experimental evaluation. Development is organised around requirements, records and document design, web application development, terminal implementation, integration and testing. The filing system is treated as the foundation of the application, while fingerprint recognition supplies identity evidence for selected departmental activities.

The methodology addresses the two research questions established in Chapter One. The first concerns the organisation, controlled updating and retrieval of structured records and uploaded documents. The second concerns the association of fingerprint evidence with the appropriate student record and its use within a reliable workflow. Separate subsystem tests and integration tests are used because a functioning scanner or backend alone does not establish the behaviour of the complete application.

## 3.2 Study Context and Requirements

### 3.2.1 Departmental context and information sources

The project concerns student filing and management in the Department of Electrical and Electronic Engineering, Obafemi Awolowo University. The project owner reported on 17 October 2026 that records are predominantly hard copies, are difficult to retrieve, and require considerable effort to review, update and correct. The owner also confirmed that the filing system includes both structured records and uploaded documents.

This clarification provides a requirements input and a local description of the problem. It is not treated as a quantified retrieval-time study or a completed programme of stakeholder interviews. Requirements were further informed by the supplied literature, the team's recorded software and attendance decisions, the existing data models and API contracts, and the documented hardware feasibility work. A fuller process description remains necessary to establish record categories, custody, indexing and departmental correction authority.

The literature was reviewed by separating project drafts from reference publications and checking whether the latter supported the claims attributed to them. Hardware descriptions were reconciled with the actual terminal records. This process informed the filing-first structure of the system and removed unsupported assumptions about recognition accuracy, component specifications and completed network integration.

### 3.2.2 Functional requirements

Table 3.1 defines the principal requirements. Implementation status refers to the development evidence available for this draft, rather than to production deployment.

**Table 3.1:** Principal system requirements and implementation position

| ID | Requirement | Implementation position |
|---|---|---|
| R1 | Maintain a stable student identity with validated profile and academic information | Backend models and controlled operations implemented. |
| R2 | Search and retrieve the permitted student and academic records | Backend APIs implemented; connected browser workflow still requires validation. |
| R3 | Upload, classify, associate and retrieve supporting documents privately | Confirmed scope; document-storage implementation remains proposed. |
| R4 | Control corrections and preserve the relevant academic or document history | Academic rules and selected history mechanisms exist; document revision history and comprehensive audit remain incomplete. |
| R5 | Enforce user permissions and adviser, course or ownership scope | Backend access mechanisms implemented for the corresponding operations. |

Continued on next page

| ID | Requirement | Implementation position |
|---|---|---|
| R6 | Acquire fingerprints and carry out the supported recognition experiments | Diagnostic, recognition-pilot and deferred-capture firmware paths implemented; their test contexts are distinct. |
| R7 | Associate an enrolled biometric identity with the permanent student record | Restricted enrolment/binding and integrated resolution remain to be completed. |
| R8 | Preserve offline captures and transfer them without losing event identity | Device storage, USB export and pending backend ingestion implemented. |
| R9 | Apply identity, roster, timing and permission checks before biometric attendance is recorded | Attendance and roster foundations exist; complete biometric reconciliation remains proposed. |
| R10 | Produce evidence of correctness, timing and selected failure behaviour | Subsystem and backend evidence exists; records-task, document and full integration evaluations remain to be completed. |

### 3.2.3 Non-functional requirements

The system requires consistent records, bounded processing, controlled disclosure and recoverable operation. Invalid or conflicting input should be rejected without leaving a partial update. A repeated submission should preserve the identity of the original operation. Missing identity or timing information should remain unresolved rather than being converted into an assumed attendance decision.

Performance is evaluated through defined timing intervals and datasets. No institutional retrieval-time target or population biometric-accuracy target has been established by the present requirements material. These targets must be agreed before any final acceptance study that depends on them. Functional acceptance can nevertheless be specified precisely: a permitted request must return the correct record, a denied request must not disclose it, and a rejected transaction must leave the protected state unchanged.

## 3.3 Development Procedure and System Architecture

### 3.3.1 Iterative development procedure

Development proceeded through successive records/backend, terminal and integration increments. The software work established student and academic entities, access policies and controlled services. Hardware work established scanner communication, reference transfer, recognition experiments and persistent unidentified capture. The later backend increment introduced authorised preview and ingestion of capture batches while keeping them pending identity and timing review.

The remaining increments concern private document management, restricted biometric identity binding, reviewed reconciliation and connected browser operation. Each increment is evaluated against its requirements before its behaviour is described as a completed system function. This approach permits faults discovered through subsystem, HTTP or database tests to inform the next revision without implying

that all development followed a single uninterrupted sequence.

### 3.3.2 Logical architecture

The records application separates browser presentation, backend services and persistent data. Django and Django REST Framework implement the current backend. Structured records are represented through Django models and migrations. PostgreSQL is the intended application database and has been used in isolated integration validation; SQLite has also been used for isolated development tests. Shared-source reconciliation and production database cutover are separate activities.

The browser interface is a client of the backend rather than a direct database client. The reviewed frontend folder contains prototypes; their demonstration state is not evidence of complete API integration. The proposed document subsystem adds metadata management and private file storage to the existing structured-record services.

Figure 3.1 distinguishes the implemented deferred-capture route from the remaining integration. Solid arrows show established logical interfaces; dashed arrows identify proposed components or connections. The diagram does not assert completed browser integration or direct device-to-server Wi-Fi transfer.

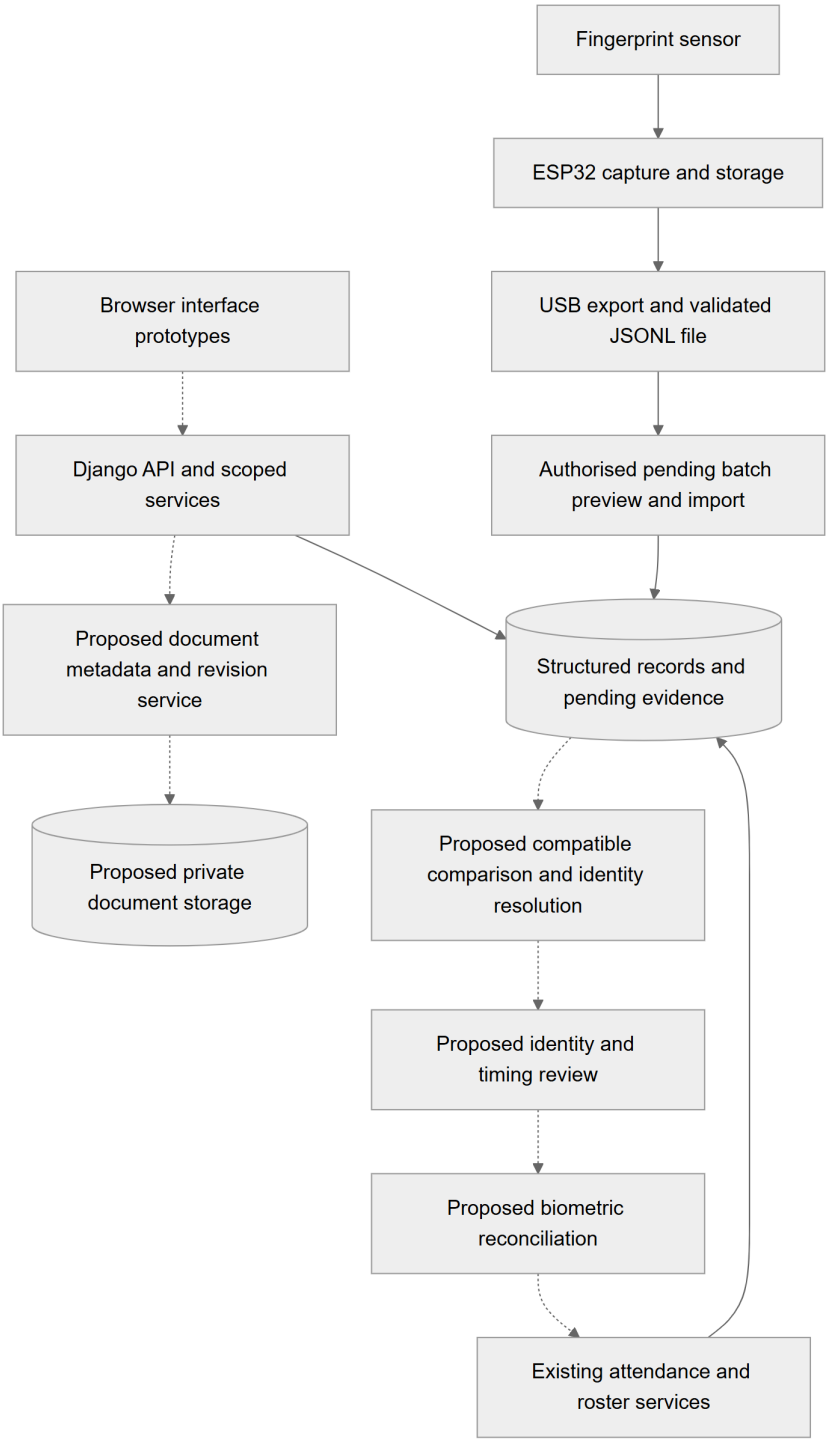



**Figure 3.1:** System architecture with implemented and proposed connections

The design keeps the identity and records responsibilities explicit. The database can store pending biometric evidence before it can interpret that evidence as a student identity. Private document storage similarly requires its own access and consistency rules rather than being implied by the presence of student tables.

## 3.4 Records and Database Design

### 3.4.1 Student identity and core entities

The implemented Student model uses a permanent UUID as its primary identifier. Matriculation or UTME information is stored as a separate textual identifier, preserving leading zeros and allowing the student's internal identity to remain stable. An optional account link associates a login user with a student profile. Guardians are separate records linked to the student.

Academic information is organised through sessions, courses, course offerings and result attempts. A course offering associates a course with a session, semester, level, credit units and grading scale. A result links a student to an offering and an attempt number. It also retains the credit units and grading reference needed for the historical calculation.

**Table 3.2:** Selected implemented records and integrity rules

| **Entity** | **Principal information** | **Important relationship or rule** |
|---|---|---|
| Student | Permanent UUID, external identifier, name, admission and current academic information | External identifier is unique; account linkage is optional and one-to-one. |
| Guardian | Name, relationship and contact information | Linked to a student; several guardian records may belong to the same student. |
| Academic Session | Academic-year label | Unique session with a validated consecutive-year format. |
| Course | Course code and title | Course code is unique. |
| Course Offering | Course, session, semester, level, units and grading scale | Course/session/semester combination is unique. |
| Result | Student, offering, attempt, score, saved units, grading scale and verification state | Student/offering/attempt combination is unique; a missing score remains distinct from zero. |
| Grading Scale | Version name, grade-point bands and publication state | Published grading versions are protected from alteration. |
| Adviser Assignment | Staff, academic session, level and active state | Controls the relevant adviser scope. |
| Course Roster and approval | Confirmed course membership and revision history | Class sessions preserve the approved membership used when they are opened. |
| Attendance Record | Class session, student, status and responsible actor | At most one record for a student in a class session. |

Continued on next page

| Entity | Principal information | Important relationship or rule |
|---|---|---|
| Pending Capture Batch and event | Class-bound metadata, immutable probe event, boot identity and uptime | Pending evidence is separate from an attendance record. |

Figure 3.2 shows the principal academic relationships. It is a selected view of the implemented schema rather than the complete database.

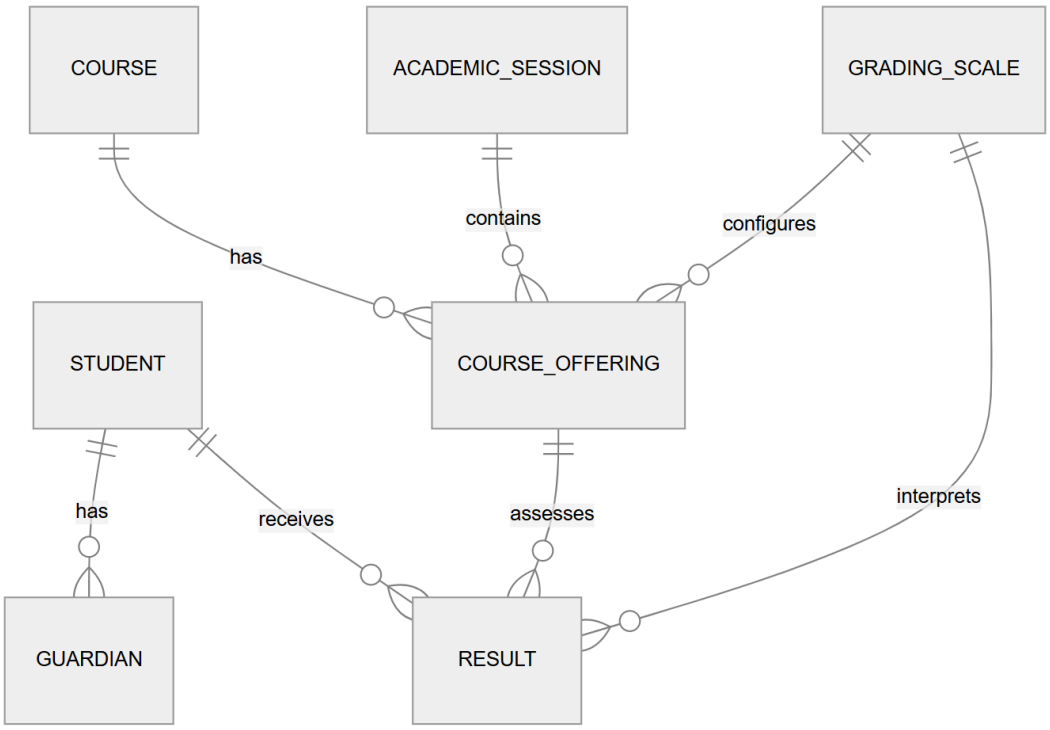



**Figure 3.2:** Selected student and academic record relationships

### 3.4.2 Validation transactions and correction

Validation is applied to the meaning and permitted range of inputs as well as their representation. Examples include valid academic periods, finite numerical scores, positive credit units and permitted changes to student information. Database constraints prevent specified duplicate identities, result attempts and attendance records. Application services add the permission and workflow checks required before a change is accepted.

Related updates use transactions where partial completion would create an inconsistent state. The current attendance and import services also use locks to coordinate competing operations. The purpose is to preserve a defined outcome when submissions, retries or permission changes occur concurrently. These mechanisms apply the constraint and transaction principles described by the PostgreSQL Global Development Group [6], [7]; the specific workflow rules are properties of the project implementation.

A correction to structured information is distinguished from replacing a supporting document. Structured changes follow the relevant field and permission rules. A replacement document should retain a link to its preceding version and a reason for the change. Comprehensive audit recording and the proposed document-history mechanisms require further implementation; the presence of selected correction records does not establish an application-wide audit trail.

### 3.4.3 Academic calculations

The backend calculates grade-point averages from verified result records using saved credit units and the applicable published grading scale. The general expression is:

$$GPA = \frac{\sum_{i=1}^{n} u_i g_i}{\sum_{i=1}^{n} u_i}, \tag{3.1}$$

where $u_i$ is the saved credit units and $g_i$ is the grade point for result $i$. Semester GPA uses the selected session and semester, while the implemented CGPA function uses the recorded cumulative attempts. Failed and repeated recorded attempts are included under the current project rule. This rule must be reconciled with authoritative departmental policy before being described as an official university calculation policy.

**Table 3.3:** Initial project grading configuration

| **Score interval** | **Grade point** |
|---|---|
| 0 up to but excluding 40 | 0 |
| 40 up to but excluding 45 | 1 |
| 45 up to but excluding 50 | 2 |
| 50 up to but excluding 60 | 3 |
| 60 up to but excluding 70 | 4 |
| 70 through 100 inclusive | 5 |

The implementation rejects invalid or missing scores in a supplied calculation set and requires verification of the supplied results. It

returns no average where there are no results. A numerical CGPA is kept separate from academic completeness and graduation review. Independent worked examples, boundary values and incomplete-history cases form part of the evaluation procedure.

## 3.5 Proposed Document Filing Subsystem

### 3.5.1 Metadata and revision design

Uploaded documents are confirmed project scope, but their private storage and revision management are not yet established as implemented. The proposed design associates every document with the permanent student UUID and separates its descriptive metadata from the stored file. It also separates the continuing document identity from individual uploaded revisions.

**Table 3.4:** Proposed document records

| **Proposed record** | **Information to retain** | **Purpose** |
|---|---|---|
| Student Document | Document UUID, student UUID, title, category, relevant academic period and current revision reference | Organises the continuing document within the student's file. |
| Document Revision | Revision UUID/sequence, document reference, original filename, internal storage key, detected file type, byte length and integrity digest | Identifies the particular uploaded content without using its public filename as an access mechanism. |

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| Proposed record | Information to retain | Purpose |
|---|---|---|
| Revision provenance | Uploader, upload time, source description and replacement/correction reason | Explains how the document entered the system and why a later revision exists. |
| Document state | Validation/review state and any approved archive or retention information | Prevents an incomplete or rejected upload being treated as an available record. |

The entity names in Table 3.4 are proposed design names, not existing Django models. Document categories, permitted formats, file-size limits, revision authority and retention periods remain configuration and policy decisions to finalise. Candidate categories can be derived from the actual departmental inventory rather than assumed to cover every type of university document.

### 3.5.2 Upload and retrieval procedure

The proposed upload workflow first authenticates the user and checks permission for the target student and document category. The uploaded file is staged while its size, detected type and content-validation requirements are checked. Descriptive metadata is validated, and an internal storage key associates the private file with the revision record. Only an accepted revision becomes available through authorised retrieval.

The design must coordinate storage and database failures. A database transaction cannot by itself make a separate filesystem write atomic. A staged upload therefore needs an explicit completion state

and a cleanup or recovery procedure for interrupted publication. The evaluation will verify that an incomplete upload is not presented as a valid document and that failures do not produce a published record with missing content.

Retrieval begins with an authorised search of student and document metadata. Before download or display, the backend rechecks permission and resolves the internal storage key. Knowing a file path or document UUID should not bypass this check. A proposed replacement creates a new revision with a reason and actor; it does not silently overwrite the earlier revision or its provenance.

The first implementation can support metadata-based retrieval without claiming OCR or full-text search. If extraction of scanned document contents is later selected, it requires an additional recognition and validation procedure. The existing PDF results parser is a separate layout-specific tool and does not establish OCR capability for arbitrary scanned files.

## 3.6 Access Control and Web Application Design

The current application uses staff and student accounts with server-side action permissions and record scope. Ordinary adviser operations require the relevant permission and an active assignment for the current academic session and student level. Attendance operations additionally use explicit course assignments. A linked student account accesses the student's own permitted information. The active HOD account is

represented through the application's privileged administration policy.

**Table 3.5:** Selected access contexts

| **User context** | **Records or activity context** | **Access condition** |
|---|---|---|
| HOD administration | Departmental configuration and permitted administrative operations | Active privileged staff identity and the relevant administrative workflow. |
| Ordinary adviser | Student profiles and selected academic operations | Action permission plus active assignment in the current academic session and level. |
| Assigned course staff | Attendance class, roster-related views and permitted recording/correction | Relevant attendance permission and active assignment to the course offering. |
| Student account | Own profile and permitted verified results or attendance information | Active account linked to that student's profile; no general staff record access. |
| Document user | A particular student's document/category | Proposed explicit document permission and scope; final category policy remains to be implemented. |

The implemented JSON API uses session authentication and CSRF controls for the relevant browser write requests. Input and permission checks occur in the backend even when the interface also provides validation. Browser screens require integration tests for authentication, loading, invalid input, denied access and successful record operations. Their final evaluation must use the connected application rather than

screenshots or prototype state alone.

Supporting functions such as messaging, guardian information and selected result-sharing operations exist in the backend. Their inclusion in the final report should remain proportional to the filing and identification objectives and to the functions actually demonstrated. They do not substitute for testing the required records and document workflows.

## 3.7 Hardware Configuration and Firmware Design

### 3.7.1 Tested components and interfaces

The hardware records identify a regular ESP32-D0WD-V3 development board with 4 MB flash and an AS608-labelled fingerprint module whose internal firmware version is unknown. The scanner reports a reference capacity of 150 and was used at security level 3 with 57,600-baud UART communication. The PC serial recording interface uses 115,200 baud. Upload baud rate, scanner baud rate and console baud rate are separate settings.

The DS1307 clock was used for the recorded recognition pilot. The later deferred-capture firmware does not integrate its calendar-time path and deliberately retains null capture dates. Optional OLED, button and feedback configurations are described by the terminal design but should not all be presented as physically validated parts of the measured pilot.

**Table 3.6:** Regular ESP32 interface configuration

| **Interface or function** | **Configuration** | **Evidence status** |
|---|---|---|
| Scanner TX to ESP32 RX | GPIO16 | Working UART connection recorded in hardware tests. |
| Scanner RX from ESP32 TX | GPIO17 | Working UART connection recorded in hardware tests. |
| Scanner UART | 57,600 baud, 8N1 | Recorded scanner communication configuration. |
| PC console/recorder | 115,200 baud | Firmware and recorder configuration. |
| I2C SDA/SCL | GPIO21/GPIO22, 100 kHz | Configured bus used by the clock path; final assembly schematic must show actual connections. |
| DS1307 address | 0x68 | Clock adapter configuration and pilot timekeeping. |
| Optional SSD1306 display | 128 × 64, configured address 0x3C | Optional adapter; inclusion in the final build needs its own evidence. |
| Enrol/ delete/ menu buttons | GPIO25/ GPIO26/ GPIO27 | Regular-board UI configuration, separate from the accuracy and deferred-capture profiles. |
| Green/red LED outputs | GPIO18/GPIO19 | Configured feedback outputs; actual parts and resistor values require final schematic evidence. |
| Buzzer control | GPIO23 | Configured output; the proposed driver stage is not established as a measured assembled circuit. |

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| Interface or function | Configuration | Evidence status |
|---|---|---|

The regular ESP32's capabilities are documented by Espressif Systems [12]; ESP32-S3 specifications are not substituted for the tested board. The DS1307 datasheet specifies a 4.5–5.5 V device supply and standard-mode I2C operation [13]. The board-side bus must remain compatible with ESP32 logic levels; breakout pull-ups and any level translation belong in the final schematic. Actual supply rail and current measurements are still needed before a measured power budget or battery runtime can be reported.

For design analysis, an LED resistor can be selected from $R = (V_{drive} - V_F)/I_F$ using the actual LED and drive limits. Average load power can be estimated from measured rail quantities as $P_{avg} = \sum_k V_k I_{k,avg}$. Where a battery is implemented, an estimated runtime is $t = E_{usable}\eta/P_{avg}$ when usable battery energy is defined before conversion. These expressions are design tools, not results from measurements already performed.

### 3.7.2 Firmware separation and capture state

The firmware uses C++ with the Arduino framework under PlatformIO. Separate build profiles support normal diagnostics, whole-gallery recognition evaluation, selected-slot evaluation and deferred unidentified capture. This separation preserves the experimental meaning of each mode. The accuracy builds do not create application attendance records,

while the deferred profile disables the older local attendance UI.

The capture controller requires finger removal between presentations and records only a completed acquisition attempt according to its defined procedure. Communication commands use bounded waits and validated responses. Uncertain persistence is reported rather than automatically retried as a destructive change.

The general deferred-capture states are shown in Figure 3.3. Storage or recovery faults require attention; rebooted capture does not silently resume without explicit arming.

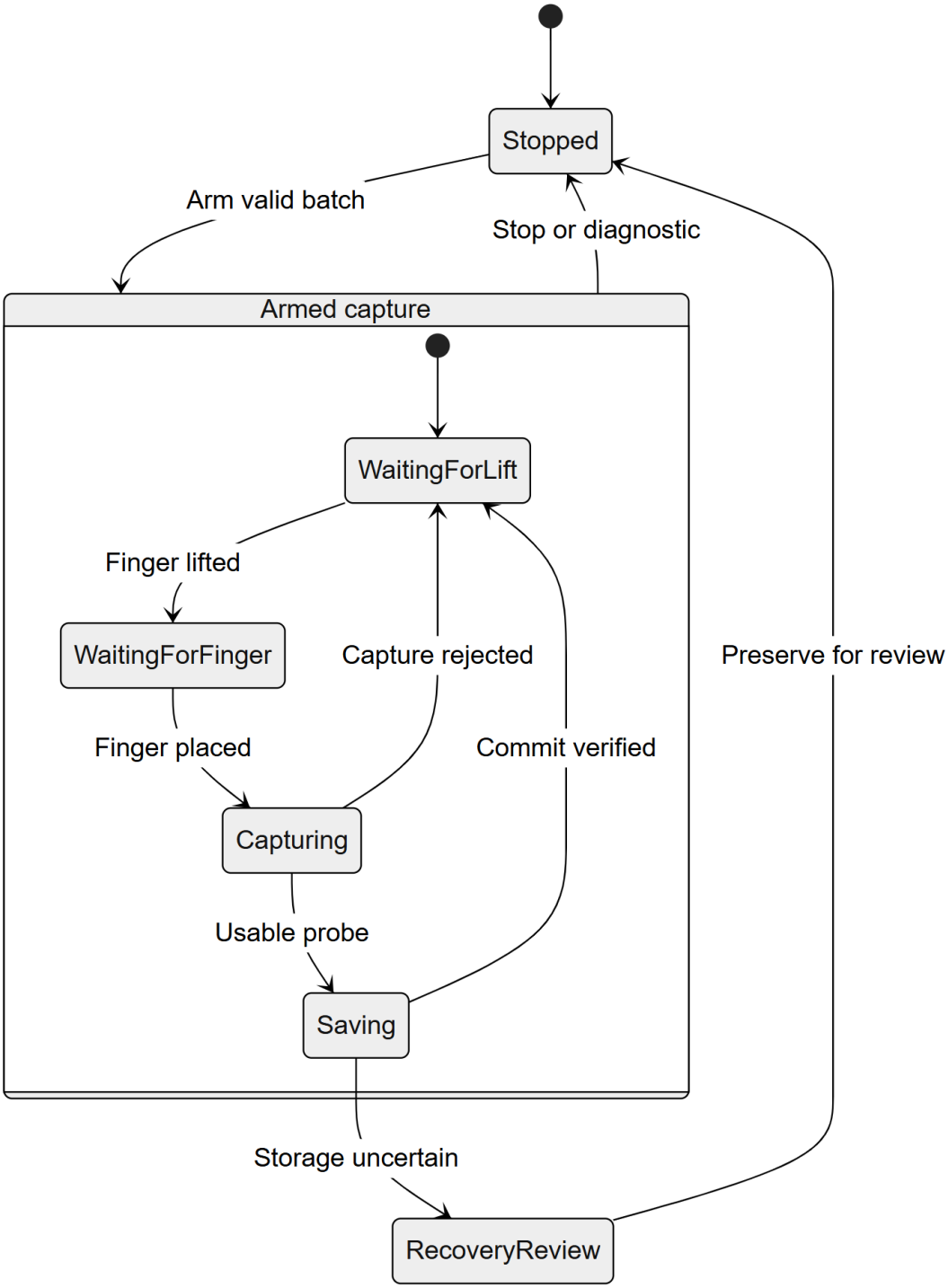



**Figure 3.3:** Simplified deferred-capture state model

The diagram is a behavioural summary. The implemented controller and recovery tests provide the detailed state transitions, timeouts and limits.

## 3.8 Fingerprint References and Identity Integration

### 3.8.1 Enrolment and reference representation

The diagnostic enrolment procedure acquires the presentations needed by the sensor to form a reference and stores it in a checked free slot. Normal diagnostic and accuracy-build enrolment use their documented procedure; the separate button UI contains an additional confirmation presentation and should not be described as the recognition pilot's enrolment protocol.

Reference transfer was implemented through the scanner's character-buffer and storage commands. The hardware log records a transferred opaque reference payload of 1,536 bytes, its saved integrity digest and readback checks. That observed length is not a universal specification for every scanner model, representation or UART packet. The project distinguishes the transferred payload from the transport packet size and from the number of occupied gallery slots.

The proposed enrolment registry associates a template/reference UUID with the student UUID, finger label, scanner/format compatibility information and gallery/slot binding. A local slot is a reference location that may change after restoration or library replacement. Exact-byte hashes detect content integrity; they are not used to recog-

nise a person or to prove unique enrolment.

### 3.8.2 Matching architecture and workflow decisions

The saved recognition experiments perform comparison using the scanner. Later saved-probe feasibility work also used scanner-assisted direct comparison. A compatible software-only server matcher has not been established for the opaque representation. The integration design therefore leaves matching location and orchestration explicit: a backend can coordinate compatible comparison without implying that Django itself decodes the template or executes the sensor's proprietary algorithm.

For a resolved biometric outcome, the proposed workflow checks the active reference-to-student binding, the student's eligibility for the selected class and the authority of the operation. Unknown, conflicting, inactive or out-of-roster outcomes remain unresolved or are rejected for the specified reason. A successful scan is not automatically an attendance record.

The attendance foundation uses course offerings, HOD-confirmed rosters and per-class membership snapshots. The candidate list can suggest active students at the relevant level and students with a latest verified failed prior attempt; a failure is not treated as proof of current registration. Reviewed roster membership, rather than course credit units, establishes the class population used by the workflow.

### 3.8.3 Selectable attendance configurations in one installed build

The regular ESP32 application provides two runtime attendance configurations in one installed SH1106-capable firmware. Recompilation and reinstallation are not required to change attendance type. The operator opens Attendance Mode, chooses Immediate or Deferred, and confirms the choice. Start/Stop and Deferred Class Batch are separate menu pages. Entering a menu stops deferred acquisition; the operator must explicitly restart it. Both acquisition paths are mutually exclusive clients of the scanner UART.

**Table 3.7:** Runtime attendance configurations in the unified terminal

| **Property** | **Immediate local matching** | **Deferred capture** |
|---|---|---|
| Identity decision | Scanner searches existing enrolled references during the presentation. | No identity decision at capture; compatible matching and review occur later. |
| Stored output | Local timestamp, reference hash, slot, score and event identifiers in CSV. | Unidentified probe and batch/event/boot identifiers in verified ESP32 files. |
| Required setup | Enrolled gallery, valid RTC and writable log storage. | Prepared class-session batch, compatible scanner and writable capture storage; no preloaded gallery required. |

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| **Property** | **Immediate local matching** | **Deferred capture** |
|---|---|---|
| Positive feedback | Attendance Saved after a verified local append; one short beep and green indication. | Capture Saved / Pending Match after commit; one short beep with both LEDs off. |
| Duplicates | Current prototype: same template and local date. Final class-session binding remains outstanding. | Finger lift separates capture attempts; resolved student/class duplicates require later reconciliation. |
| Runtime selection | Choose Immediate and confirm; Start/Stop controls acquisition. | Choose Deferred and confirm; select a prepared batch and explicitly start it. |
| Persistence | Existing CSV history is retained across mode changes. | Committed files and selected batch are retained; reboot leaves acquisition stopped. |
| Reference slots | Attendance uses volatile capture buffers and existing references; it creates no new enrolled slot. | Attendance uses volatile capture buffers and ESP32 files; it creates no enrolled slot. |
| Website status | Local save is not a website attendance submission. | Import is pending evidence; it is not confirmation that a named student attended. |

In immediate mode, a fresh presentation is compared against the scanner's enrolled references. A valid DS1307 timestamp and a verified local append are required before the terminal reports Attendance Saved. This checkpoint records local check-ins and suppresses a repeated template on the same local date. It does not yet implement the final student/class-session identity reconciliation; the daily rule must not be

described as a per-class attendance rule.

Deferred mode stores the unidentified feature probe on the ESP32 and reports Capture Saved / Pending Match only after commit and readback checks. It does not search or add a reference to scanner flash. A backend class-session batch is prepared through the USB tool; the physical interface can select a prepared batch and start or stop capture without reflashing or maintaining a laptop connection during acquisition. The selected batch is retained across restart, but capture is never automatically armed after a reboot. Calendar timing and identity reconciliation retain the limitations described in Section 3.9.

Neither attendance configuration consumes a new permanent scanner reference slot per presentation. Only enrolment or an explicit reference deployment writes such slots. Starting or changing attendance mode does not erase reference templates, old check-ins or pending probes. First-time student registration remains a separate integration requirement: the enrolment template must pass through the laptop bridge to an authorised backend student binding, with success acknowledged only after the database commit. A sensor-only Enrolled message does not establish that website registration has completed.

Feedback distinguishes verified local outcomes from pending evidence. Green is reserved for confirmed sensor enrolment, a verified immediate check-in append, or a verified deliberate fingerprint deletion. Startup, progress, mode selection, clock configuration, export and deferred capture do not light green. In particular, the rule concerns a verified operation, not whether enrolment has occurred since the

most recent boot: restored references can also produce valid immediate matches. Appendix B specifies the event feedback, confirmation behaviour and remaining integration states.

### 3.9 Deferred Capture Transfer and Pending Ingestion

The deferred design captures fingerprints without requiring a preloaded per-class reference gallery. The terminal stores each unidentified probe in a class-bound batch with stable device, batch, event and boot identifiers. Event timing is retained as boot-relative elapsed milliseconds. These values preserve sequence and timing evidence but are not converted into an invented calendar date.

The device uses LittleFS and staged writes. It writes pending content, closes and reads it back, commits it under a new filename and verifies the committed content before reporting a saved capture. CRC32 fields detect corruption in the binary records. An interrupted pending write blocks further capture until its condition is reviewed; valid committed data can be exported with a recovery flag.

USB export transfers a bounded stream to a PC, where the downloader checks count and integrity information and creates a JSONL interchange file. The device files are retained after download. Neither a successful download nor a backend receipt authorises deletion of the source captures.

**Table 3.8:** Implemented interchange bounds and meanings

| **Property** | **Implemented bound or rule** |
|---|---|
| Capture probe | Maximum 2,048 decoded bytes per event. |
| Batch | Maximum 512 capture events. |
| Backend import file | Maximum 2 MiB. |
| JSONL line | Maximum 4,096 bytes. |
| Append behaviour | New snapshots may append to the exact stored prefix; existing events retain content and order. |
| Exact retry | Recovers the durable receipt for the accepted file snapshot. |
| Conflict | Changed event content, reused event identity or inconsistent batch binding rejects the transaction. |
| Capture calendar time | Null until trusted timing provision and review are implemented. |
| Import effect | Pending evidence only; no automatic student identity, attendance, absence or warning. |

The backend provides authorised preview, import and metadata-listing operations. Preview checks the complete file without persisting it. Import revalidates permission, class binding, file digest, schema and conflicts before committing evidence and a durable receipt atomically. A preview digest binds the selected bytes; it does not replace authentication or prove the physical device's identity.

The proposed reconciliation stage follows ingestion. It requires a compatible identity-resolution procedure and explicit timing review before invoking an attendance operation. Until this stage is implemented, the captures remain pending. Device authentication, encrypted biomet-

ric storage and key management, retention rules and final operational hardening also remain unfinished.

## 3.10 Software and Records Evaluation Procedure

### 3.10.1 Evaluation levels

Evaluation is organised into record/service tests, database and HTTP integration tests, terminal experiments and complete user workflows. These levels address different questions. A unit or service test checks a defined operation and its resulting state. Database contention tests examine interactions between simultaneous operations. HTTP tests exercise authentication, request handling and response behaviour. Browser and physical integration tests establish behaviour beyond those isolated components.

The current evidence includes backend validation in isolated databases and separate hardware experiments. Document-management, connected browser tasks and complete biometric reconciliation require additional evaluation. The numerical outcomes of completed tests belong in Chapter Four; the procedures and scope are specified here.

### 3.10.2 Implemented backend validation

Backend tests use synthetic student and academic fixtures rather than live departmental records. The documented checks include record permissions, result calculations, validation, attendance services and pending capture ingestion. Database-specific checks were run against

an isolated PostgreSQL 16.15 instance. This environment was separate from the shared departmental source database.

The pending-ingestion validation included competing transactions using independent connections. A coordinator checked that one transaction was genuinely waiting on a database lock before the competing operation proceeded. Cases covered identical imports, competing append/prefix snapshots, conflicting event reuse, permission revocation, session changes and rollback. This procedure tests defined consistency properties under bounded contention; it is not a sustained throughput benchmark.

The documented HTTP cases used actual login, CSRF, session cookies and the original JSONL request bytes. They included permitted ingestion, anonymous or invalid-CSRF requests, revoked access, interrupted uploads, retry after a discarded response, and a fresh application-process restart. These checks extend beyond forced-authentication service tests while remaining separate from browser interaction and field deployment.

### 3.10.3 Proposed records and document task study

The records-task evaluation will use a declared synthetic dataset containing distinct students, academic periods, valid and invalid result cases, and representative supporting documents. The dataset size and document categories must be recorded when the study is performed. Examples will include similar student names, missing information, duplicate identifiers and documents with replacement revisions so that

success requires returning the correct information rather than merely any search result.

Each task will specify the requested information, expected student/document identity and permitted user context. A retrieval task ends when the correct required information is displayed or the correct document is opened. A correction task additionally checks the saved state and the retained history or verification requirements. Valid and invalid inputs will be evaluated separately.

Document tests will cover accepted and rejected files, metadata association, permitted download, denied direct access, revision replacement and interruption during upload publication. Tests of storage/database failure will verify that incomplete content is not exposed as a completed record. Archive and retention behaviour will be evaluated only after their policy and implementation have been established.

If a comparison with hard-copy filing is conducted, the manual and digital conditions will use equivalent record sets and lookup requests. The study will record operator preparation, starting conditions, completion rules and task order. Where the same operator performs both conditions, order and familiarity effects will be considered. Until this study is performed, no percentage retrieval improvement or measured workload reduction is claimed.

## 3.11 Fingerprint Recognition Pilot Procedure

### 3.11.1 Participants references and experimental conditions

A functional recognition pilot was conducted on 15 October 2026 using one participant, one enrolled finger and one designated un-enrolled finger. The scanner contained one freshly enrolled reference in slot 1. Both recognition modes used that same reference at security level 3 and the recorded scanner communication settings. Finger labels were supplied by the operator; they were not independently determined from the recorder output.

The pilot was designed to check operation under these conditions. It did not sample a departmental student population or establish identification among several enrolled identities. Repeated presentations of the same fingers are correlated observations and are reported as repeated attempts, not independent participants.

**Table 3.9:** Recognition pilot design

| **Property** | **Whole-gallery experiment** | **Selected-slot experiment** |
|---|---|---|
| Firmware profile | esp32-accuracy-identify | esp32-accuracy-verify |
| Scanner operation | Search over the reported reference range | Search restricted to the selected occupied slot |
| Operating mode | 1:N identification | Claimed-reference 1:1 check using selected-slot Search |

Continued on next page

| Property | Whole-gallery experiment | Selected-slot experiment |
|---|---|---|
| Reference gallery | One enrolled template, slot 1 | Same enrolled template, slot 1 |
| Genuine attempts | Ten new presentations of the enrolled finger | Ten new presentations of the enrolled finger |
| Negative attempts | Ten presentations of the designated unenrolled finger | Ten presentations of the designated different finger |
| Trial separation | Detected lift and minimum two-second cooldown | Detected lift and minimum two-second cooldown |

### 3.11.2 Trial collection

Before each run, the reference inventory and settings were checked and the intended genuine identity selected for the recorder. A genuine block and a negative block were then collected under the operator-declared ground truth. The gallery remained fixed during the run. The expected identity in the identification experiment labelled the correct result without limiting the gallery search.

Each automatic trial used a new presentation, one feature-generation operation and one Search transaction. A continuously held finger did not generate repeated presentation trials. The recorder captured trial identifiers, declared case, returned identity where available, decision, processing errors, raw accepted-match score and timing information. Invalid or unavailable score values were retained as unavailable rather than replaced with a fabricated zero.

The two firmware modes were run separately, and their observations were analysed with separate denominators. Earlier diagnostic captures and later saved-probe comparisons were not pooled into these runs. In particular, the selected-slot Search pilot was not relabelled as a statistical direct-Match experiment.

### 3.11.3 Classification of outcomes

For a genuine identification attempt, a returned identity matching the expected slot was classified as a correct acceptance. No match was a genuine failure, and an accepted wrong identity was also a genuine identification failure with an additional wrong-identity flag. A negative search returning an enrolled identity was a false positive. A completed negative search returning no match was a correct rejection.

In the selected-slot experiment, the selected reference defined the claimed identity. A contradictory identity outside that requested range was treated as a protocol error rather than a valid verification result. Acquisition, communication, sensor and unclassified errors were recorded separately from completed decisions.

The automatic recorder observes detected acquisition attempts. It does not independently prove how many physical placements were entirely undetected. Polling errors without an established presentation were retained separately, and the absence of a separate physical-placement ledger limits claims about total presentation coverage.

### 3.11.4 Timing and calendar checks

Two processing intervals were retained. The total processing interval began with the image-acquisition command and ended with the completed recognition result. Search time covered the Search transaction alone. Human waiting, the inter-trial cooldown and the subsequent RTC read were excluded from these processing intervals. These measurements therefore differ from a complete attendance transaction or an operator's record-retrieval task.

The DS1307 was synchronised from the PC separately for each run using Lagos local time, UTC+01:00. Trial-linked calendar readings and additional clock-status checks were recorded. Calendar sequence and differences from PC receipt time were inspected as functional checks. Serial and host delay, PC clock error and one-second calendar resolution prevent those differences from being treated as calibrated absolute clock accuracy or long-term drift.

## 3.12 Reference Transfer and Deferred Capture Experiments

### 3.12.1 Reference transfer and restoration

Reference portability was investigated separately from recognition accuracy. An enrolled reference was exported from the scanner, saved with its metadata and integrity digest, and restored through the supported buffer/storage commands. Readback was compared with the original bytes. The procedure distinguishes successful byte transfer from successful recognition and from compatibility with another physical

scanner or software matcher.

The hardware record also documents an explicitly authorised erase/restore experiment after the occupied library and backup coverage had been checked. This historical procedure is reported as an experiment, not as a routine requirement for enrolment or evaluation. Scanner slots and gallery state were recorded so that a changed library could not be treated as the same experimental reference set without qualification.

### 3.12.2 Saved-probe persistence and later comparison

On 16 October 2026, the deferred firmware was used to acquire a live probe and preserve it in ESP32 storage under a synthetic class-session context. The batch was downloaded and validated, then downloaded again after an operator-reported USB power cycle. The saved outputs were compared for byte equality and integrity.

The saved probe was subsequently compared against the enrolled reference through scanner-assisted direct comparison. No new finger presentation was taken during that final comparison. This procedure evaluates preservation and subsequent usability of the saved representation. It is separate from the 15 October repeated-presentation study and is not a software-only server-matching experiment.

The physical check did not interrupt an unfinished flash write under a controlled brownout. Native storage and controller tests modelled interrupted-write and reboot states, but those tests are not described as measured electrical power-failure behaviour. The distinction is retained

when discussing recovery results.

## 3.13 Performance Measures and Analysis

### 3.13.1 Records and document measures

For a defined set of assigned record tasks, task correctness is calculated as:

$$Task\ correctness = \frac{N_{correctly\ completed}}{N_{assigned}} \times 100\%. \tag{3.2}$$

A task is correctly completed only when the expected student information or document is returned and the required state is preserved. The test log records failures and denied operations separately so that a fast incorrect retrieval cannot count as successful performance.

For timing observations $t_i$, the elapsed interval is $t_i = t_{finish,i} - t_{start,i}$. User-task time, HTTP response time and database-operation time are labelled separately. Repeated task times will be summarised with the observation count, median and range or a suitable percentile. Document sizes and network conditions accompany upload/download measurements where they affect the interval.

If a comparable manual baseline is available, an indicative median time reduction can be expressed as:

$$Time\ reduction = \frac{\widetilde{t}_{manual} - \widetilde{t}_{digital}}{\widetilde{t}_{manual}} \times 100\%, \tag{3.3}$$

provided the manual median is positive and the tasks and timing

rules are comparable. This expression defines a planned analysis; no baseline value or improvement percentage has yet been supplied.

### 3.13.2 Biometric decision measures

The biometric analysis uses the trial definitions in Section 3.11 and the firmware's separate error counters. Let $TP$, $TN$, $FP$ and $FN$ denote completed decision outcomes. For genuine identification attempts, $FN$ includes both no-match failures and wrong-identity decisions; the wrong-identity count is a subset and is not added a second time. Let $N_{recorded}$ include all recorded attempts, including processing errors, and let:

$$N_{decision} = TP + TN + FP + FN. \tag{3.4}$$

The following measures are then reported where their denominators are non-zero:

$$Decision\ accuracy = \frac{TP + TN}{N_{decision}} \times 100\%, \tag{3.5}$$

$$\begin{aligned} Sensitivity\ or\ recall &= \frac{TP}{TP + FN} \times 100\%, \\ Specificity &= \frac{TN}{TN + FP} \times 100\%, \end{aligned} \tag{3.6}$$

$$Decision\ coverage = \frac{N_{decision}}{N_{recorded}} \times 100\%. \tag{3.7}$$

Acquisition, communication, sensor and unclassified-error counts are also reported against the recorded-attempt denominator. They

are not silently removed from the description of overall operation merely because completed-decision accuracy excludes them. A zero denominator produces an unavailable measure, not a zero error rate.

For the whole-gallery protocol, completed un-enrolled searches returning an enrolled identity define the false-positive identification rate:

$$FPIR = \frac{FP}{FP + TN} \times 100\%. \tag{3.8}$$

Completed genuine searches returning no match or the wrong identity define the stated top-one identification failure measure:

$$FNIR = \frac{FN}{TP + FN} \times 100\%. \tag{3.9}$$

These are protocol-defined identification rates, kept separate from the selected-slot decision rates. Verification/identification and threshold-dependent error terminology follow the distinction discussed by Fiumara [11] and the biometric literature reviewed in Chapter Two.

The raw sensor matching score is retained only as supplementary comparison evidence. It is not converted into a probability, and accepted-only scores are not used to construct an unsupported equal-error rate or complete ROC/DET curve. Zero observed errors are reported as counts over the observed attempts rather than proof of a zero underlying error probability.

### 3.13.3 Integration and reliability measures

Integration evaluation checks whether each accepted operation creates the intended state and whether rejected or repeated inputs preserve it. Capture-import cases verify event counts, immutability, rollback and receipt identity. Once biometric reconciliation is implemented, each resolved event will be traced through identity binding, eligibility and timing checks to the final attendance operation.

**Table 3.10:** Evaluation groups and required evidence

| **Test group** | **Expected observation** | **Position in the present evidence** |
|---|---|---|
| Student identity and record integrity | Correct associations; prohibited duplicates and invalid references rejected | Existing backend tests; final evidence set to identify in Chapter Four. |
| Academic calculations | Agreement with independent examples and correct handling of boundaries/missing data | Implemented calculation tests; institutional policy reconciliation remains separate. |
| Access scope | Permitted requests succeed and prohibited requests disclose or change no protected information | Existing backend tests; browser interaction still requires evaluation. |
| Document upload and publication | Valid content attached to the intended student; incomplete/invalid uploads not published | Proposed subsystem and tests. |

Continued on next page

| Test group | Expected observation | Position in the present evidence |
|---|---|---|
| Document retrieval and revisions | Authorised retrieval, denied direct access and preserved replacement history | Proposed subsystem and tests. |
| Recognition pilot | Per-trial correct/wrong identity, rejection, error and timing evidence | Recorded one-participant pilot with separate recognition modes. |
| Reference/ probe preservation | Byte integrity across transfer, restoration or the stated restart procedure | Separate recorded hardware experiments. |
| Pending ingestion and replay | Immutable evidence, atomic conflict rejection and durable retry receipts | Existing database and HTTP validation. |
| Biometric-to-attendance reconciliation | Correct student/class association; no premature or duplicate attendance effect | Proposed remaining integration. |
| Complete filing tasks and manual comparison | Correct user-task completion with consistently defined time | Planned; dataset and baseline need collection. |

Failure analysis records the initial condition, attempted operation, observed failure, resulting state and any subsequent correction. A passing synthetic regression, a physical terminal observation and a field result are distinct kinds of evidence. None is used as a substitute for an unperformed test at another level.

## 3.14 Data Handling and Methodological Limits

Initial software tests used synthetic fixtures. Biometric trial records and saved representations are retained in restricted local working storage and are not required in the public report. The report presents aggregate outcomes and non-sensitive metadata needed to understand the experiment. Document and biometric access, retention and eventual deletion require explicit policies as the system progresses towards operational use.

The present recognition study is limited by one participant, one enrolled reference and repeated presentations of two designated fingers. It does not test multiple-identity gallery selection or population error rates. Timing comparisons with other studies are also limited by different operation boundaries, scanners, datasets and retry procedures. The later saved-probe experiment establishes a different feasibility question and is discussed separately.

Additional volunteer studies will require a documented recruitment and consent procedure and the appropriate project approvals before data collection. Those procedures are not retroactively claimed for the current pilot. Live departmental source-data use and deployment are likewise separate from isolated synthetic validation.

The final power arrangement, physical enclosure fit, full document workflow, trusted timing for deferred captures, device authentication and complete biometric reconciliation remain development or evaluation work. These limits define the claims that can be made from the current

prototype; they do not remove the corresponding requirements from the approved project scope.

## 3.15 Traceability to the Project Objectives

**Table 3.11:** Relationship between objectives methods and report evidence

| **Objective from Chapter One** | **Main methodology sections** | **Evidence for Chapter Four** |
|---|---|---|
| Analyse the filing workflow and requirements | 3.2 and 3.3 | Dated local process evidence and requirements/access matrix. |
| Design the records document and identity model | 3.4, 3.5 and 3.8 | ER/data dictionary, document design and tested integrity rules. |
| Implement controlled web records and document functions | 3.4–3.6 and 3.10 | Connected operations, document cases, access results and filing-task observations. |
| Implement the fingerprint terminal functions | 3.7, 3.8, 3.11 and 3.12 | Firmware/configuration, hardware evidence and defined recognition/preservation experiments. |

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| Objective from Chapter One | Main methodology sections | Evidence for Chapter Four |
|---|---|---|
| Integrate biometric identity with the selected workflow | 3.8, 3.9 and 3.13 | Identity binding, pending ingestion and final reconciliation cases, with unfinished links identified. |
| Evaluate records biometrics and integration | 3.10–3.14 | Separate functional, timing, biometric and integration outcomes with limitations. |

## 3.16 Chapter Summary

The methodology translates the filing problem into structured records, supporting-document requirements, controlled web operations and a defined biometric integration path. It describes the implemented backend and terminal work, the recorded recognition and preservation procedures, and the proposed work needed for document management and complete identity reconciliation. The evaluation separates records correctness, biometric decisions and integration behaviour so that Chapter Four can report what each part demonstrates and where evidence remains incomplete.

# CHAPTER 4

# IMPLEMENTATION RESULTS AND DISCUSSION

**Completion outline—not final results.** No Chapter Four draft was included in the supplied report-review Markdown set. The structure below follows the materials review; populate it from the actual, dated evidence records.

## 4.1 Implementation Checkpoint and Evaluation Scope

Identify the software revision, terminal build, datasets, participants and test conditions used for each evaluation. Keep implemented, proposed and untested functions distinct.

## 4.2 Student Records and Document Workflows

Present expected-versus-observed backend, browser, record-integrity and access results. Report document upload, retrieval and revision outcomes only where implemented and tested. Add the actual filing-task dataset and any comparable manual baseline, with task counts and consistent timing definitions.

## 4.3 Terminal Implementation and Assembly

Add the actual circuit schematic, component values, verified connections, supply measurements and assembly photographs. Put diagrams in folder/ and photographs in plates/; label and cite each figure or plate in the text.

## 4.4 Fingerprint Recognition Results

Use the documented pilot's trial records. Separate whole-gallery identification and selected-slot verification denominators, wrong identities, rejections, processing errors and timing observations. Do not merge the later saved-probe feasibility experiment into the pilot dataset.

## 4.5 Reference Preservation and Pending Capture

Present byte-integrity, restoration, capture persistence, USB transfer and pending ingestion evidence separately. Pending ingestion alone is not completed identity-to-attendance reconciliation.

## 4.6 Integration Results and Limitations

Report completed identity binding, roster, permission, timing, retry and duplicate handling cases, identifying any remaining links explicitly. Compare findings to the literature only where the evaluated conditions and metrics are comparable.

# CHAPTER 5

# CONCLUSIONS AND RECOMMENDATIONS

**Completion outline—not final conclusions.** No Chapter Five draft was included in the supplied Markdown set. Complete this chapter after Chapter Four has an agreed evidence checkpoint.

## 5.1 Summary of the Work

Summarise the departmental filing problem, records/document design, fingerprint subsystem and actual integration scope.

## 5.2 Answers to the Research Questions

Answer each of the two research questions from Chapter One using the results reported in Chapter Four. Identify which objectives were achieved, partly achieved or remain outstanding.

## 5.3 Limitations

State limitations of the implemented scope, reference compatibility, identity binding, participant/sample counts, test conditions and external validity.

## 5.4 Recommendations for Further Work

Prioritise remaining document management, biometric reconciliation, connected user-workflow evaluation and any approved deployment work. Do not introduce new measurements or unsupported claims of improved performance here.

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# APPENDICES

# APPENDIX A

# SUPPORTING EVALUATION EVIDENCE

**Supporting material pending.** Add de-identified trial protocols, test-case summaries and suitably redacted evidence that supports Chapter Four. The source-notes directory contains the original draft evidence index and materials review for the team's use; those internal notes are not automatically included as submitted academic material.

Do not insert raw fingerprint captures, biometric references, real student records or credentials into a shared report project. Use approved, de-identified evidence and cite the dataset or test record associated with each result.

# APPENDIX B

# SUPPLEMENTARY DESIGN MATERIAL

## B.1 Terminal Feedback and Operator Controls

The following rules refine the proposed scanner feedback schedule and extend it to both attendance configurations and maintenance operations. The schedule forms the implemented software contract. Host simulations and firmware builds validate the control logic; physical mode switching and assembled LED/buzzer timings still require bench confirmation. These checks do not constitute an end-to-end website attendance validation. All indications are transient. Green or red indications last 700 ms; a short operation beep is 100 ms, a key click is 20 ms and an enrolment-completion beep is 700 ms. Error and retry signals contain two 120 ms beeps separated by 120 ms. Deliberate deletion contains two 100 ms beeps separated by 100 ms. The scheduler is nonblocking and key clicks do not interrupt an active success/error sequence.

**Table B.1:** Defined feedback for terminal events

| Event | Screen or result | Green | Red | Buzzer |
|---|---|---|---|---|
| Startup | Starting, then Home; attendance stopped | Off | Off | One short beep |
| Enrolment attempt 1 accepted | Lift Finger, then Enrol 2 of 3 | Off | Off | None |
| Enrolment attempt 2 accepted | Lift Finger, then Enrol 3 of 3 | Off | Off | None |
| Enrolment complete | Enrolled; verified scanner slot | On | Off | One long beep |
| Unreadable enrolment scan | Try Again; lift finger first | Off | On | Two error beeps |
| Enrolment mismatch, full gallery or unconfirmed save | Specific failure; no success claim | Off | On | Two error beeps |
| Already enrolled | Already Enrolled; existing slot | Off | On | Two error beeps |
| Immediate attendance saved | Attendance Saved; local record saved | On | Off | One short beep |
| Already marked | Already Checked In; same template/date | Off | On | Two error beeps |
| Not recognised or unreadable attendance | No Match or Scan Failed; lift and retry | Off | On | Two error beeps |
| Match without valid clock/storage | Matched / Not Saved; reason displayed | Off | On | Two error beeps |
| Deferred capture saved | Capture Saved / Pending Match | Off | Off | One short beep |
| Deferred capture unreadable | Try Again; lift and reposition | Off | On | Two error beeps |
| Deferred communication, storage, integrity or capacity fault | Capture Stopped; files retained | Off | On | Two error beeps |
| Held finger | Lift between scans; no extra record | Off | Off | None |

Continued on next page

| Event | Screen or result | Green | Red | Buzzer |
|---|---|---|---|---|
| Missing class batch | Capture Not Started; prepare class via USB | Off | On | Two error beeps |
| Mode confirmation | Use Immediate/Deferred? Existing records kept | Off | Off | Key click only |
| Menu held after USB maintenance | Mode selection; capture remains stopped | Off | Off | One short beep |
| Mode confirmed/start succeeds | Selected mode running | Off | Off | One short beep |
| Physical stop | Selected mode stopped | Off | Off | One short beep |
| Batch prepared/selected | Batch prepared; class UUID available | Off | Off | One short beep |
| Menu/cancel | Return without saving or deleting | Off | Off | Click; cancel gives one short beep |
| Confirmation or action timeout | Cancelled, or Action Timed Out | Off | Off/On | Cancel beep, or two error beeps |
| Delete awaiting confirmation | Identify, lift, then fresh Delete press | Off | Off | Key click only |
| Delete or full-library erase verified | Deleted / Library Erased; history kept | On | Off | Two deletion beeps |
| Delete/erase unconfirmed | Unconfirmed; inspect library | Off | On | Two error beeps |
| Clock saved | Clock Saved; readback verified | Off | Off | One short beep |
| Clock save failed | Clock Not Saved; check DS1307 | Off | On | Two error beeps |
| Log export complete | Log Exported; USB output complete | Off | Off | One short beep |
| Deferred export complete | Batch Exported; device files retained | Off | Off | One short beep |
| Export failure | Specific failure; evidence retained | Off | On | Two error beeps |
| Diagnostics, backup or restore | Attendance Paused; console protocol results | Off | Off | Silent during transfers |

Continued on next page

| Event | Screen or result | Green | Red | Buzzer |
|---|---|---|---|---|
| Confirmed USB filesystem format | Console completion/failure; all local files affected | Off | Off | Silent maintenance operation |
| Background missing clock, sensor or storage | Home readiness warning | Off | Off | No repeated alarm; attempted operation reports failure |
| Button press | Current screen retained unless action changes it | Off | Off | Short click; active outcome feedback has priority |

Green indicates a verified local persistent biometric operation, not a generic acknowledgement and not a server-side identity guarantee. Deferred captures remain unidentified even when stored correctly. A duplicate immediate check-in does not add another row. Export success confirms the local stream operation, not a durable laptop file or backend receipt. USB protocol maintenance suppresses automatic capture and uses console-specific results without incidental green indications or unsolicited feedback inside transfer frames.

## B.2 Mode Selection and Recovery

The installed application starts with attendance stopped. On Attendance Mode, Enrol selects Immediate and Delete selects Deferred; a fresh Enrol press confirms the selection and starts the chosen path when its prerequisites are satisfied. Menu cancels. A confirmation expires after 20 s. No mode selection erases data. The Start/Stop page changes acquisition state without changing attendance type.

Deferred Class Batch cycles through valid batches prepared by the USB tool and shows the selected class-session UUID. The terminal retains the selection but requires explicit re-arming after restart. An interrupted or damaged batch is not silently repaired, replaced or reformatted. An unavailable or full capture store stops capture and retains existing evidence. Existing scanner diagnostic, backup and restore commands pause acquisition until an explicit restart.

After USB maintenance, a fresh two-second Menu hold opens attendance-mode selection with capture still stopped. The UI handles this hold only after the active serial transaction has completed, preserving framed transfers. The deferred statistics page counts pending captures rather than identified students; its export page directs the operator to the laptop batch downloader. Unassigned Delete presses on Start/Stop and Class Batch pages do not initiate deletion.

### B.3 States Reserved for Incomplete Integration

Automatic student-enrolment upload and database binding remain unfinished. The future laptop/website interface must distinguish Sending Enrolment, Registration Pending and Registration Saved. Only a durable database acknowledgement may produce the last state; sensor-only enrolment is insufficient. Likewise, deferred matching requires separate Matched, Unknown, Ambiguous and Out-of-roster review states before a student/class attendance write. These integrations remain unfinished.

## B.4 Remaining Assembly Material

The final circuit, resistor values and measured driver-stage behaviour still require assembly evidence. The existing conceptual and methodology diagrams are included in Chapters Two and Three; their editable sources remain in folder/.